NEEL VITTAL BHARATH VADLAMUDI

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SUMMARY

Data Scientist completing an M.P.S. in Data Science at UMBC (GPA 3.67, May 2026) with a B.Tech in Computer Science from NIT Delhi and 19+ months of combined professional and research experience in data engineering, machine learning, and AI validation for regulated, data-intensive environments. Built production ETL pipelines processing **500K+ daily records** across 8 source systems with automated data integrity safeguards, deployed cloud infrastructure on **Azure Databricks at 99.7% uptime**, and collaborated with **Meta AI Research** on transformer model evaluation that improved accuracy by 15%. Track record of delivering cross-functional analytics, automated quality monitoring, and reproducible research in environments where compliance and audit-readiness matter.

TECHNICAL SKILLS

DATA ENGINEERING & CLOUD: Python, SQL, R, PySpark, Apache Spark, Azure Databricks, Azure Data Factory (foundational), Azure Data Lake (foundational), AWS S3/Lambda (foundational), ETL/ELT Pipelines, REST APIs, Flask, Kubernetes, CI/CD, Git

AI/ML & NLP: PyTorch, HuggingFace Transformers, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent AI, LSTM, Ensemble ML (XGBoost, LightGBM), Anomaly Detection, A/B Testing, Causal Inference

ANALYTICS & VISUALIZATION: Power BI, Tableau, Monte Carlo Simulation, VaR/CVaR Risk Modeling, Statistical Modeling, Process Monitoring, Pandas, NumPy, GeoPandas, Matplotlib

DATA INTEGRITY & COMPLIANCE: Automated validation and reconciliation workflows, cross-system data quality monitoring, reproducible research methodology, audit-traceable evaluation frameworks, GxP-aware data pipeline design

EXPERIENCE

Data Infrastructure Research Assistant — May 2025 - Aug 2025

AI Institute of South Carolina — *3 months*

- Engineered Python and Apache Spark ETL pipelines consolidating **2.3M+ records from 5 agencies** into a unified data lakehouse architecture, reducing data preparation time by **70%** and enabling reproducible downstream analytics for cross-functional research teams.
- Deployed and maintained Azure Databricks cloud infrastructure with Kubernetes orchestration at **99.7% uptime**, supporting 4 PhD researchers running concurrent analytical workloads across validated data environments.

Machine Learning Research Assistant — Dec 2024 - Feb 2025

University of Maryland, Baltimore County (Meta AI Collaboration) — *3 months*

- Fine-tuned transformer architectures using PyTorch and HuggingFace, building rigorous evaluation frameworks with documented benchmarking protocols that improved model accuracy by **15%** through reproducible experimental design.
- Built automated data collection and transformation pipelines consolidating distributed training outputs into standardized, audit-ready datasets, cutting reporting turnaround from **5 days to 8 hours**.

Business Analyst, Data Engineering and Risk Analytics — Jun 2023 - Jul 2024

HDFC Bank — *13 months*

- Architected ETL pipelines integrating **8 source systems processing 500K+ daily records** with automated validation, cross-system reconciliation, and data integrity monitoring that prevented **$2.1M in downstream analytical failures**.
- Built Monte Carlo simulation and VaR/CVaR risk models running **1M+ daily scenarios**, delivering probabilistic dashboards in Power BI to senior leadership across **4 business lines** for strategic decision-making.
- Automated **15+ analyst-hours/week** of manual reporting and reduced reconciliation cycle time by **60%** through Python-driven quality monitoring and continuous improvement of data workflows.

EDUCATION

M.P.S. Data Science, University of Maryland, Baltimore County (UMBC) — Expected May 2026 | GPA: 3.67

B.Tech Computer Science and Engineering, National Institute of Technology Delhi — May 2023

KEY PROJECTS

CDC Epidemiological Intelligence Platform | *RAG + LLMs + LSTM + Apache Spark*

Public health analytics platform combining Retrieval-Augmented Generation with time-series forecasting on CDC datasets. Natural language queries against epidemiological data with automated trend detection, outbreak pattern recognition, and traceable analytical outputs.

Multi-Agent AI Prediction Platform | *LLMs + REST API + Python*

Multi-agent system where specialized LLM agents collaborate through structured, auditable workflows to generate validated analytical predictions with full reasoning traceability and REST API integration for downstream consumption.

Real-Time IoT Process Monitoring Pipeline | *LSTM + XGBoost + Streaming Data*

Continuous environmental monitoring pipeline processing real-time sensor data through ensemble models for automated anomaly detection, predictive alerting, and threshold-based escalation workflows.

Cloud Operations Analytics Platform | *Azure Databricks + Power BI + ML*

Databricks-based platform processing 700MB+ operational datasets with ML-driven classification models and Power BI dashboards for cross-functional KPI monitoring, trend analysis, and data-driven decision support.